

A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-90903207})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K , the normalized generator σ_x , the pair (a', J) representing the torsion class e_j , the auxiliary divisor I' , the compactly represented element t , a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1 - \sigma_x)^2 I' + \text{div}(t) + i_x(J) = 0, \quad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$ -basis of the ideal. For example, the ideal $J = \begin{pmatrix} 673 & 535 \\ 0 & 1 \end{pmatrix}$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}673 + \mathbf{Z}(535 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\text{Cl}(K)$ (as HNF ideals). A character x on $\text{Cl}(K)/5$ is recorded by its value vector $(x(\bar{\kappa}_1), x(\bar{\kappa}_2), x(\bar{\kappa}_3))$; thus $(1, 0, 0)$ is the character dual to $\bar{\kappa}_1$. The column number j records which basis element e_j of $\text{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x . Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x .

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 - s + 22725802 = 0$, of discriminant -90903207 . Class group invariants $(110 \ 10 \ 5)$, class number 5500; the three stored generator ideals (HNF) are $\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 896 & 877 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 3394 & 3238 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a , column e_1

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (-s + 357)y^3 + (38s - 114069)y^2 + (-635s + 4146437)y + (-170s - 54121585)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 + 717y^8 - 229526y^7 + 31601333y^6 - 1933306401y^5 \\ & + 77857898333y^4 - 2073339028520y^3 + 38405453887314y^2 - 4438833 \\ & 22121305y + 2929811939259475 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{11600637}{138062368994593} + \left(-\frac{392}{138062368994593}\right)s, \quad J = \begin{pmatrix} 673 & 535 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 673$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $12725614073817 = 3^3 \cdot 471319039771$:

$$\begin{pmatrix} 1413957119313 & 348818173482 & 1114084523112 & 1352294157834 & 342096363317 & 824465606609 & 156655225808 & 110565819356 & 1235078961937 & 496652078670 \\ 0 & 3 & 0 & 0 & 2 & 0 & 1 & 2 & 1 & 1 \\ 0 & 0 & 3 & 0 & 1 & 1 & 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 7 factors with signed exponents of absolute value up to 2; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 3 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 2: character a , column e_2

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (-s + 357)y^3 + (38s - 114069)y^2 + (-635s + 4146437)y + (-170s - 54121585)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 + 717y^8 - 229526y^7 + 31601333y^6 - 1933306401y^5 \\ & + 77857898333y^4 - 2073339028520y^3 + 38405453887314y^2 - 4438833 \\ & 22121305y + 2929811939259475 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{23913437}{5260658229025227} + \left(\frac{746}{5260658229025227}\right)s, \quad J = \begin{pmatrix} 2163 & 142 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2163$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $12725614073817 = 3^3 \cdot 471319039771$:

$$\begin{pmatrix} 1413957119313 & 348818173482 & 1114084523112 & 1352294157834 & 342096363317 & 824465606609 & 156655225808 & 110565819356 & 1235078961937 & 496652078670 \\ 0 & 3 & 0 & 0 & 2 & 0 & 1 & 2 & 1 & 1 \\ 0 & 0 & 3 & 0 & 1 & 1 & 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 7 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (0 \ 3 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 3: character a , column e_3

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (-s + 357)y^3 + (38s - 114069)y^2 + (-635s + 4146437)y + (-170s - 54121585)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 4y^9 + 717y^8 - 229526y^7 + 31601333y^6 - 1933306401y^5 \\ + 77857898333y^4 - 2073339028520y^3 + 38405453887314y^2 - 4438833 \\ 22121305y + 2929811939259475$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{659205707}{450359356492424224} + \left(-\frac{26359}{450359356492424224}\right)s, \quad J = \begin{pmatrix} 3394 & 3238 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3394$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $9469881199371 = 3 \cdot 3156627066457$:

$$\begin{pmatrix} 9469881199371 & 7161191189476 & 7085545526483 & 8409893984811 & 9413668750897 & 6552318643806 & 1854373582889 & 4900672185672 & 7818925049375 & 2559453208712 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 7 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 3 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 4: character b , column e_1

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4y^4 + (-s + 801)y^3 + (-4s + 146024)y^2 + (645s - 1438566)y + (-1872s - 55615392)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 8y^9 + 1617y^8 + 285640y^7 + 19321939y^6 + 315860928y^5 \\ - 9487944165y^4 - 541251861120y^3 - 4379027249088y^2 + 1050995603 \\ 49696y + 3172815876243456$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{11600637}{138062368994593} + \left(-\frac{392}{138062368994593}\right) s, \quad J = \begin{pmatrix} 673 & 535 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 673$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2012190851484 = 2^2 \cdot 3^5 \cdot 11 \cdot 188195927$:

$$\begin{pmatrix} 12420931182 & 5518516710 & 1217295990 & 7244974677 & 10554051375 & 2822054093 & 10234598128 & 10778534868 & 1045305715 & 9700512824 \\ 0 & 3 & 0 & 0 & 0 & 1 & 2 & 0 & 2 & 1 \\ 0 & 0 & 3 & 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 0 & 0 & 0 & 3 & 1 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 3 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 150 factors with signed exponents of absolute value up to 390153445668; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 5: character b , column e_2

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4y^4 + (-s + 801)y^3 + (-4s + 146024)y^2 + (645s - 1438566)y + (-1872s - 55615392)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 8y^9 + 1617y^8 + 285640y^7 + 19321939y^6 + 315860928y^5 \\ & - 9487944165y^4 - 541251861120y^3 - 4379027249088y^2 + 1050995603 \\ & 49696y + 3172815876243456 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{23913437}{5260658229025227} + \left(\frac{746}{5260658229025227}\right) s, \quad J = \begin{pmatrix} 2163 & 142 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2163$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $17320006900416 = 2^6 \cdot 3 \cdot 11^5 \cdot 560123$:

$$\begin{pmatrix} 1626597192 & 1468253204 & 150269740 & 1150889212 & 1382772184 & 713437330 & 1345522113 & 1575711433 & 1082871654 & 137899727 \\ 0 & 242 & 193 & 88 & 4 & 65 & 138 & 94 & 238 & 16 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 22 & 20 & 1 & 7 & 2 & 2 & 21 \\ 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 155 factors with signed exponents of absolute value up to 368821611190; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 0 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 6: character b , column e_3

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4y^4 + (-s + 801)y^3 + (-4s + 146024)y^2 + (645s - 1438566)y + (-1872s - 55615392)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 8y^9 + 1617y^8 + 285640y^7 + 19321939y^6 + 315860928y^5 \\ & - 9487944165y^4 - 541251861120y^3 - 4379027249088y^2 + 1050995603 \\ & 49696y + 3172815876243456 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{659205707}{450359356492424224} + \left(-\frac{26359}{450359356492424224}\right)s, \quad J = \begin{pmatrix} 3394 & 3238 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3394$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $122784686044416 = 2^8 \cdot 3^4 \cdot 11 \cdot 17^3 \cdot 109567$:

$$\begin{pmatrix} 16719047664 & 1975027836 & 1065226170 & 2091891192 & 14106478437 & 6010757004 & 15099545454 & 2660213807 & 3069441170 & 6567327435 \\ 0 & 102 & 30 & 36 & 51 & 58 & 9 & 95 & 6 & 74 \\ 0 & 0 & 3 & 0 & 1 & 0 & 2 & 0 & 2 & 0 \\ 0 & 0 & 0 & 24 & 13 & 17 & 7 & 14 & 8 & 13 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 155 factors with signed exponents of absolute value up to 12865992887; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (3 \ 0 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 7: character c , column e_1

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4y^4 - 351y^3 + (-16s + 29802)y^2 + (208s - 4320840)y + (16448s + 29478272)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 8y^9 - 686y^8 + 62396y^7 - 8756623y^6 + 72623492y^5 + 95 \\ & 02752388y^4 - 429426473408y^3 + 9447602402368y^2 - 99308430893568 \\ & *y + 7017616947560448 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{11600637}{138062368994593} + \left(-\frac{392}{138062368994593}\right)s, \quad J = \begin{pmatrix} 673 & 535 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 673$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $62418872678400 = 2^{11} \cdot 3^5 \cdot 5^2 \cdot 7 \cdot 716707$:

$$\begin{pmatrix} 19265084160 & 6029996130 & 14573151825 & 1615094022 & 15933457212 & 10530512397 & 8625296352 & 5375110570 & 12220580154 & 18233629617 \\ 0 & 30 & 15 & 24 & 0 & 14 & 17 & 28 & 13 & 10 \\ 0 & 0 & 3 & 0 & 0 & 0 & 1 & 2 & 0 & 2 \\ 0 & 0 & 0 & 6 & 0 & 3 & 2 & 4 & 5 & 2 \\ 0 & 0 & 0 & 0 & 6 & 5 & 5 & 5 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 226 factors with signed exponents of absolute value up to 333270424482; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \ 2 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 8: character c , column e_2

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4y^4 - 351y^3 + (-16s + 29802)y^2 + (208s - 4320840)y + (16448s + 29478272)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 8y^9 - 686y^8 + 62396y^7 - 8756623y^6 + 72623492y^5 + 95 \\ & 02752388y^4 - 429426473408y^3 + 9447602402368y^2 - 99308430893568 \\ & *y + 7017616947560448 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{23913437}{5260658229025227} + \left(\frac{746}{5260658229025227} \right) s, \quad J = \begin{pmatrix} 2163 & 142 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2163$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $13461303389184 = 2^{10} \cdot 3^3 \cdot 7 \cdot 69554519$:

$$\begin{pmatrix} 17527738788 & 16275929212 & 537995854 & 14471558808 & 11874756348 & 1241162132 & 6962415374 & 533604480 & 12029995317 & 10675394590 \\ 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 6 & 2 & 2 & 4 & 4 & 1 & 5 \\ 0 & 0 & 0 & 0 & 4 & 0 & 2 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 4 & 0 & 3 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 221 factors with signed exponents of absolute value up to 495178176932; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \ 3 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 9: character c , column e_3

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4y^4 - 351y^3 + (-16s + 29802)y^2 + (208s - 4320840)y + (16448s + 29478272)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 8y^9 - 686y^8 + 62396y^7 - 8756623y^6 + 72623492y^5 + 95 \\ & 02752388y^4 - 429426473408y^3 + 9447602402368y^2 - 99308430893568 \\ & *y + 7017616947560448 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{659205707}{450359356492424224} + \left(-\frac{26359}{450359356492424224}\right)s, \quad J = \begin{pmatrix} 3394 & 3238 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3394$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $622375914674304 = 2^7 \cdot 3^4 \cdot 7 \cdot 8575505879$:

$$\begin{pmatrix} 1440684987672 & 34252957320 & 344180541252 & 1430544082658 & 665874173682 & 803579986447 & 728210031805 & 506178240890 & 168047728331 & 354151960872 \\ 0 & 12 & 0 & 3 & 0 & 5 & 8 & 10 & 7 & 7 \\ 0 & 0 & 12 & 4 & 3 & 1 & 7 & 10 & 4 & 7 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 1 & 0 & 2 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 230 factors with signed exponents of absolute value up to 6694573388851; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 1 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 3401y^3 + (-45s - 61044)y^2 + (465s - 4980174)y + (13392s + 47669877)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} + 6802y^8 - 122133y^7 + 1606918y^6 - 320021187y^5 + 1587530 \\ & 3883y^4 - 18563558472y^3 - 3500133396633y^2 - 191813227477479y + \\ & 6348829617810441 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ 1 \ -1 \ 1 \ -1 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{11600637}{138062368994593} + \left(-\frac{392}{138062368994593}\right)s, \quad J = \begin{pmatrix} 673 & 535 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 673$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $324410514353559 = 3^8 \cdot 41 \cdot 24097 \cdot 50047$:

$$\begin{pmatrix} 445007564271 & 308680969104 & 249262503192 & 432378669465 & 8813331765 & 259193366509 & 383759122093 & 116393320134 & 267547266960 & 43868885602 \\ 0 & 9 & 0 & 3 & 0 & 1 & 6 & 8 & 2 & 0 \\ 0 & 0 & 9 & 0 & 0 & 1 & 0 & 2 & 3 & 0 \\ 0 & 0 & 0 & 3 & 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 0 & 3 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 566 factors with signed exponents of absolute value up to 788275807797126485514; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_1) = (0 \ 0 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 3401y^3 + (-45s - 61044)y^2 + (465s - 4980174)y + (13392s + 47669877)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} + 6802y^8 - 122133y^7 + 1606918y^6 - 320021187y^5 + 1587530 \\ & 3883y^4 - 18563558472y^3 - 3500133396633y^2 - 191813227477479y + \\ & 6348829617810441 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ 1 \ -1 \ 1 \ -1 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{23913437}{5260658229025227} + \left(\frac{746}{5260658229025227} \right) s, \quad J = \begin{pmatrix} 2163 & 142 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2163$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $35966359904463 = 3^5 \cdot 461 \cdot 1097 \cdot 292673$:

$$\begin{pmatrix} 444029134623 & 168034694727 & 410738647212 & 828149637 & 81325220199 & 303226736764 & 7097437813 & 229662447121 & 367474967190 & 162334466191 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 3 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 577 factors with signed exponents of absolute value up to 920874362386795255491; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (2 \ 3 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 3401y^3 + (-45s - 61044)y^2 + (465s - 4980174)y + (13392s + 47669877)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} + 6802y^8 - 122133y^7 + 1606918y^6 - 320021187y^5 + 1587530 \\ & 3883y^4 - 18563558472y^3 - 3500133396633y^2 - 191813227477479y + \\ & 6348829617810441 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ 1 \ -1 \ 1 \ -1 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{659205707}{450359356492424224} + \left(-\frac{26359}{450359356492424224}\right)s, \quad J = \begin{pmatrix} 3394 & 3238 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3394$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $25350282685575 = 3^3 \cdot 5^2 \cdot 39293 \cdot 955793$:

$$\begin{pmatrix} 563339615235 & 31909046205 & 428426494329 & 255401352753 & 375637722716 & 494008992530 & 136544952316 & 270432280855 & 105047096078 & 198896620463 \\ 0 & 5 & 3 & 2 & 4 & 2 & 3 & 0 & 0 & 1 \\ 0 & 0 & 3 & 0 & 2 & 2 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 3 & 2 & 1 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 2; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (0 \ 0 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (s + 262)y^3 + (24s + 129615)y^2 + (75s + 1637544)y + (2034s + 7323237)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 + 529y^8 + 258204y^7 + 25551363y^6 + 1166990841y^5 \\ & + 34132524471y^4 + 602656235370y^3 + 6927113426958y^2 + 309217678 \\ & 17327y + 147664883721339 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ -1 \ 0 \ 0 \ 1 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{11600637}{138062368994593} + \left(-\frac{392}{138062368994593}\right)s, \quad J = \begin{pmatrix} 673 & 535 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 673$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $901177749216 = 2^5 \cdot 3^2 \cdot 2081 \cdot 1503647$:

$$\begin{pmatrix} 18774536442 & 18059618250 & 17063296464 & 12299518994 & 12619287786 & 8081837136 & 7294084140 & 747208492 & 3874020284 & 669837582 \\ 0 & 2 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 6 & 2 & 0 & 3 & 0 & 1 & 5 & 4 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 7 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_1) = (2 \ 1 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (s + 262)y^3 + (24s + 129615)y^2 + (75s + 1637544)y + (2034s + 7323237)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 + 529y^8 + 258204y^7 + 25551363y^6 + 1166990841y^5 \\ & + 34132524471y^4 + 602656235370y^3 + 6927113426958y^2 + 309217678 \\ & 17327y + 147664883721339 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ -1 \ 0 \ 0 \ 1 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{23913437}{5260658229025227} + \left(\frac{746}{5260658229025227} \right) s, \quad J = \begin{pmatrix} 2163 & 142 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2163$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $901177749216 = 2^5 \cdot 3^2 \cdot 2081 \cdot 1503647$:

$$\begin{pmatrix} 18774536442 & 18059618250 & 17063296464 & 12299518994 & 12619287786 & 8081837136 & 7294084140 & 747208492 & 3874020284 & 669837582 \\ 0 & 2 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 6 & 2 & 0 & 3 & 0 & 1 & 5 & 4 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (2 \ 1 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (s + 262)y^3 + (24s + 129615)y^2 + (75s + 1637544)y + (2034s + 7323237)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 + 529y^8 + 258204y^7 + 25551363y^6 + 1166990841y^5 \\ & + 34132524471y^4 + 602656235370y^3 + 6927113426958y^2 + 309217678 \\ & 17327y + 147664883721339 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ -1 \ 0 \ 0 \ 1 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{659205707}{450359356492424224} + \left(-\frac{26359}{450359356492424224}\right)s, \quad J = \begin{pmatrix} 3394 & 3238 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3394$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $21406235190729 = 3 \cdot 17^2 \cdot 24690005987$:

$$\begin{pmatrix} 1259190305337 & 771631666323 & 817612348020 & 1004432335548 & 730042912528 & 874207267467 & 1070663323203 & 894740804307 & 53338463781 & 548546532485 \\ 0 & 17 & 1 & 13 & 4 & 13 & 4 & 2 & 6 & 3 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (3 \ 2 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s + 602)y^3 + (5s - 115482)y^2 + (220s + 636744)y + (544s + 9567040)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2y^9 + 1204y^8 - 232162y^7 + 24592271y^6 - 348318940y^5 \\ + 4651307932y^4 - 110306595104y^3 - 580516628672y^2 + 17625609739 \\ 776y + 98258841772032$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ -1 \ 0 \ 1 \ 1 \ -1 \ -1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{11600637}{138062368994593} + \left(-\frac{392}{138062368994593}\right)s, \quad J = \begin{pmatrix} 673 & 535 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 673$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $57982493392896 = 2^{18} \cdot 3 \cdot 73728553$:

$$\begin{pmatrix} 7077941088 & 7031309552 & 5057279344 & 4006051400 & 2736104252 & 6719377740 & 6521232414 & 3572812626 & 5409477834 & 3140639326 \\ 0 & 16 & 0 & 10 & 14 & 2 & 14 & 15 & 12 & 5 \\ 0 & 0 & 16 & 14 & 0 & 2 & 2 & 10 & 10 & 9 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 346 factors with signed exponents of absolute value up to 44819004878; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 2 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s + 602)y^3 + (5s - 115482)y^2 + (220s + 636744)y + (544s + 9567040)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2y^9 + 1204y^8 - 232162y^7 + 24592271y^6 - 348318940y^5 \\ + 4651307932y^4 - 110306595104y^3 - 580516628672y^2 + 17625609739 \\ 776y + 98258841772032$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ -1 \ 0 \ 1 \ 1 \ -1 \ -1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{23913437}{5260658229025227} + \left(\frac{746}{5260658229025227}\right) s, \quad J = \begin{pmatrix} 2163 & 142 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2163$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $65102805882112 = 2^8 \cdot 35753 \cdot 7112909$:

$$\begin{pmatrix} 1017231341908 & 803902129056 & 496726163974 & 736394910314 & 94518748456 & 916198534922 & 286853387180 & 813221468902 & 133898644251 & 95777554658 \\ 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 379 factors with signed exponents of absolute value up to 148486256644; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2 \ 4 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s + 602)y^3 + (5s - 115482)y^2 + (220s + 636744)y + (544s + 9567040)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 + 1204y^8 - 232162y^7 + 24592271y^6 - 348318940y^5 \\ & + 4651307932y^4 - 110306595104y^3 - 580516628672y^2 + 17625609739 \\ & 776y + 98258841772032 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ -1 \ 0 \ 1 \ 1 \ -1 \ -1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{659205707}{450359356492424224} + \left(-\frac{26359}{450359356492424224}\right) s, \quad J = \begin{pmatrix} 3394 & 3238 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3394$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $65102805882112 = 2^8 \cdot 35753 \cdot 7112909$:

$$\begin{pmatrix} 1017231341908 & 803902129056 & 496726163974 & 736394910314 & 94518748456 & 916198534922 & 286853387180 & 813221468902 & 133898644251 & 95777554658 \\ 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 351 factors with signed exponents of absolute value up to 60778112507; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \ 4 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.